

AfriCred - Technical Roadmap

Technical adaptation of the official strategic roadmap "From Web3 Vaults MVP to the African Yield Engine" (Base-first MVP -> Private launch -> Regional scale -> Pan-African infrastructure).

Status legend:

- **[Done]** deployed or implemented
- **[Partial]** partially delivered, see notes
- **[Todo]** not started

Trajectory overview

Phase	Horizon	Objective	Key product	Status
0. Final scoping	Week 0	Lock down the perimeter	Base-first MVP	[Done]
1. MVP build	Business days 1-42	Build the private platform	Web3 Vaults MVP	[Done]
2. Private MVP launch	Months 2-3	Test with selected investors	SME Credit Vault	[Todo]
3. Product-Market Fit	Months 3-6	Validate repayment, risk and impact	SME Credit + Productive Assets pilot	[Todo]
4. Controlled public launch	Months 6-12	Open to more investors	2 capped vaults	[Todo]
5. Regional expansion	Year 2	Deploy across UEMOA/ECOWAS	Multi-country vaults	[Todo]
6. Pan-African infrastructure	Year 3	Become the African Yield Engine	API, specialised vaults, institutional capital	[Todo]

Current position: Phases 0 and 1 delivered. The contract core and the investor dApp are validated on testnet, with an AgriCo Senegal pilot vault deployed on Base Sepolia. Ready for Phase 2 (private MVP on mainnet) once the hardening workstreams (emergency pause, multisig, external audit) are finalised.

Guiding principles

- Base as the sole settlement chain for the MVP; multi-chain remains a progressive extension.
- USDC as the main settlement and reporting asset.
- SME Credit Vault as the first public product; Productive Assets Vault as a pilot; SME Equity Vault in private access.
- Controlled access via allow-list (`depositorWhitelist`) on the vaults that require it.
- Operational transparency: investor dashboard, admin dashboard, impact dashboard and periodic reporting.
- Ramp gated by repayments, security audit and quality of field data.

Phase 0 - Final MVP scoping

Horizon: Week 0 **Status:** [Done] **Objective:** Lock down product, technical, treasury and security decisions before any development.

Priority actions

- [Done] Confirm Base as the sole settlement chain for the MVP (Base Sepolia today, Base mainnet in Phase 2).
- [Done] Adopt USDC as the main asset (MockUSDC on testnet, Circle USDC on mainnet via HelperConfig).
- [Done] Define vault parameters for the MVP: maxDeposits (global cap), whitelistEnabled (optional allow-list), tranching (senior/junior tranching), repaymentType (bullet / interest periodic / amortizing). Minimum ticket, per-wallet caps and lock-up are scheduled as Phase 2 extensions.
- [Done] Launch the SME Credit Vault first (AfriCredVault with repaymentType 0/1/2). Productive Assets Vault and SME Equity Vault remain in the Phase 3+ backlog.
- [Done] Technical dependencies: Base Sepolia RPC, deployment wallets, Vercel hosting, domains, IPFS pinning (Pinata). Safe multisig planned for mainnet go-live in Phase 2.

Deliverables

- [Done] Functional and technical specifications (af-contracts/README.md , af-frontend/README.md , af-frontend/public/architecture.svg).
- [Done] Risk register: clean slither report after the CEI fix on _settleCustody + manual review of the 78 tests.
- [Done] On-chain role matrix: factory owner, admin (loan NFT holder), allocator. Off-chain roles (treasury, management, investors) will be managed via the Safe multisig in Phase 2.
- [Done] Go for the MVP build.

KPIs

Indicator	Target	Status
MVP scope	100% validated	[Done]
Vault parameters	Validated per product	[Done]
Technical dependencies	Access available	[Done]

Gating decision

[Done] Product, treasury, security and data model rules are approved. Phase 1 build is authorised.

Phase 1 - Build the Web3 Vaults MVP

Horizon: Business days 1 to 42 **Status:** [Done] **Objective:** Deliver a private platform that supports wallet connection, vault deposits, position tracking and redemption workflows.

Priority actions

- [Done] Wallet onboarding via the injected connector (MetaMask, Rabby, Phantom) with auto-switch to Base Sepolia on connect. Coinbase Smart Wallet and WalletConnect can be added on demand.
- [Done] Base-first contracts: ERC-4626 (AfriCredVault)
- Caps (maxDeposits , adjustable via setMaxDeposits)
- Depositor whitelist (whitelistEnabled + setDepositorWhitelist)
- Senior/junior tranching with first-loss buffer
- Full lifecycle Closed -> Funding -> Custody -> OpenWithdrawal
- NAV frozen during custody
- On-chain restructure and recovery
- LayerZero OFTCore integrated (peers off by default, lock/unlock model)

[Done] Dashboards:

- Investor: / , /vaults , /vault/[address] (deposit/redeem, position, IPFS dossier, testnet faucet)
- Admin: /admin (3-step Create Loan wizard, Manage Loan tab, lifecycle, custody, restructure, recovery, hide-from-/vaults toggle)
- Borrower: /borrow (loan application form, IPFS pinning)

[Done] Security and quality:

- 78 forge tests (unit + integration via Lifecycle.t.sol)
- GitHub Actions CI (forge fmt / build --sizes / test)
- Slither static analysis, clean after the CEI fix on _settleCustody
- Documentation: READMEs, ROADMAP, architecture diagram in SVG/PNG/Mermaid

[Done] Transactional UX: one-click Approve+Deposit and Approve+Repay flows, pending tx hash surfaced to the user, 120s timeout on waitForTransactionReceipt , revert detection via receipt.status , chain-mismatch banner with manual fallback.

Deliverables

Deliverable	Status	Notes
Working investor dApp	[Done]	Next.js 14 on Vercel; live
Vault contracts deployed on testnet	[Done]	Factory, LoanRegistryNFT, VaultDeployer, Router, 1 pilot vault (AgriCo Senegal) on Base Sepolia (0x027c0dA455141827a76B5Aa2Dda2e85FB222001C)
Operational dashboards	[Done]	Investor + Admin complete for the Phase 1 scope; Treasury and Impact dashboards planned for Phase 3-5
Serverless backend	[Done]	Vercel routes /api/ipfs , /api/ipfs/list , /api/loan-request
Contracts + frontend documentation	[Done]	READMEs, ROADMAP, ARCHITECTURE diagram

KPIs

Indicator	Target	Status
Critical tests (cap, whitelist, lifecycle, tranching, recovery)	Covered	[Done]
Critical security incident	0	[Done]
Documentation	Available before go-live	[Done]

Gating decision

[Done] Testnet validated, end-to-end demo successful (vault creation, LP deposit, custody, repayments, settlement). Phase 2 transition authorised subject to the hardening listed in the 90-day plan.

Phase 2 - Private MVP launch

Horizon: Months 2 to 3 **Objective:** Test the system with a limited number of investors, prudent caps and real but controlled allocations.

Priority actions

- [Todo] Launch a private beta with selected investors via the vault's `depositorWhitelist`.
- [Todo] Activate the SME Credit Vault in limited production on Base mainnet.
- [Todo] Set a pilot TVL of 50,000 to 250,000 USD with strict tickets and caps. Add `minDeposit` and `maxDepositPerWallet` to the `AfriCredVault` contract.
- [Todo] Fund 5 to 10 pilot operations: working capital, inventory, agricultural campaign or trade finance.
- [Todo] Test the full chain in production: deposit, allocation, tracking, repayment, impact reporting.
- [Todo] Produce a first monthly performance, risk and impact report.

Deliverables

- [Todo] Private beta live (mainnet, allow-list).
- [Todo] First SME Credit allocations (5 to 10 real loans).
- [Todo] Incident and exception log.
- [Todo] Monthly investor and management report.
- [Todo] Repayment process validated in production.

KPIs

Indicator	Target
Deposits without critical incident	95% +
Pilot repayment rate	95% +
Impact report	Published monthly

Gating decision

Cautious cap increases only if deposits, allocations and first repayments are under control.

Phase 3 - Product-Market Fit validation

Horizon: Months 3 to 6 **Objective:** Prove that investors understand the product, that SMEs use the funds correctly and that risk controls work.

Priority actions

- [Todo] Gradually raise the caps as repayments perform (the on-chain `setMaxDeposits` lever is already in place; the policy needs definition).
- [Todo] First Productive Assets pilot: cold rooms, storage, solar irrigation or processing equipment. Requires either extending the `LoanRegistryNFT` schema or introducing a new `ProductiveAssetNFT`.
- [Todo] Standardised allocation sheets: beneficiary, amount, maturity, risk, collateral, repayment source. Currently a free-text `description` field + IPFS dossier.
- [Todo] Internal committee: risk, treasury, tech and impact.
- [Todo] External smart-contract audit (Spearbit / Cantina / Sherlock) before any significant ramp.
- [Todo] Formalise off-chain contracts with funded beneficiaries and field partners.

Deliverables

- [Todo] Standardised beneficiary sheets.
- [Todo] Productive Assets pilot activated.
- [Todo] Governance committee operational.
- [Todo] Strengthened reporting framework.
- [Todo] External audit plan.

KPIs

Indicator	Target
TVL	500,000 to 1,000,000 USD
Actors financed	25 to 50
Productive assets	1 to 3 pilots
Arrears > 30 days	< 5%
Repayment rate	95% +

Gating decision

Transition to the controlled public launch only if repayment, reporting and governance are robust enough.

Phase 4 - Controlled public launch

Horizon: Months 6 to 12 **Objective:** Turn the private MVP into a credible commercial platform, opened progressively to a broader investor base.

Priority actions

- [Todo] Gradually open access to more eligible investors (lift the manual allow-list).

- [Todo] Activate two public, capped vaults: SME Credit Vault and Productive Assets Vault.
- [Todo] Keep the SME Equity Vault in private access until legal structuring, valuation, liquidity and reporting are adapted.
- [Todo] Field partners: cooperatives, aggregators, exporting SMEs, cold-chain operators, mobile-money fintechs.
- [Todo] Quarterly reports on performance, risk, defaults, repayments and impact.
- [Todo] Partnerships: Base, Circle, Polygon/trails.build, DePIN/cold chain, impact investors.

Deliverables

- [Todo] Platform open in a controlled way.
- [Todo] Two capped commercial vaults.
- [Todo] Standard quarterly report.
- [Todo] Documented field partnerships.
- [Todo] Strengthened audit file.

KPIs

Indicator	Target
TVL	2 to 5 M USD
SMEs/actors financed	100 to 250
Productive assets	5 to 15
Active countries	Mali + Côte d'Ivoire or Senegal
Repayment	94% +

Gating decision

Move to regional expansion only if the platform demonstrates stability, repeatable reporting, portfolio quality and a field pipeline.

Phase 5 - Regional expansion and progressive multi-chain

Horizon: Year 2 **Objective:** Make AfriCred a UEMOA/ECOWAS regional infrastructure with multiple countries, multiple partners and progressive multi-chain capability.

Priority actions

- [Todo] Extend operations to 3-5 priority countries: Mali, Côte d'Ivoire, Senegal, Ghana or Benin.
- [Todo] Enable multi-chain rails only after the Base-first model is validated: Configure LayerZero peers between Base and Arbitrum/Optimism/Ethereum.
- Deploy RemoteShareInventory per remote chain to manage redemption liquidity.
- Dedicated audit of the OFT configuration before enabling.

[Todo] Integrate new liquidity sources: stablecoin issuers, DeFi funds, impact funds, family offices, diaspora capital.

[Todo] Strengthen on-chain/off-chain reporting: allocation proofs (allocator signatures), repayment proofs (oracle), impact proofs (EAS).

[Todo] Mobile-money integration for local cash-in/cash-out (Wave, Orange Money, MTN rails, etc.).

[Todo] Risk governance by country, sector, beneficiary, asset and partner.

Deliverables

- [Todo] Country playbook.
- [Todo] Progressive multi-chain architecture (Base + 1 to 2 L2s).
- [Todo] Auditable impact reporting (OnChainImpactRegistry + EAS).
- [Todo] Regional partner network.
- [Todo] Mobile-money model prepared.

KPIs

Indicator	Target
TVL	10 to 20 M USD
Countries covered	3 to 5
Actors financed	1,000 +
Productive assets	50 +
Blockchain partners	3 to 5 ecosystems

Gating decision

Move to pan-African infrastructure stage after validation of country replication, portfolio quality and institutional capacity.

Phase 6 - Pan-African productive finance infrastructure

Horizon: Year 3 **Objective:** Position AfriCred as the African Yield Engine: recognised infrastructure for financing real African assets at scale.

Priority actions

- [Todo] Launch an institutional programme for DFIs, banks, impact funds, family offices and foundations.
- [Todo] Create specialised vaults: agriculture, cold chain, trade finance, solar productive assets, SME working capital. Requires MultiLoanPortfolioVault and per-sector variants.
- [Todo] Integrate oracles, automated scoring, ESG reporting, proof of impact and on-chain attestations.
- [Todo] Build an AfriCred API so fintechs and partners can plug their SME portfolios in.
- [Todo] Prepare tokenised governance only after real traction: utility, staking, governance, incentives and reputation.
- [Todo] Position AfriCred as the reference African use case for stablecoins, RWA, DePIN and impact finance.

Deliverables

- [Todo] Institutional programme.
- [Todo] AfriCred API.
- [Todo] Sector-specialised vaults.
- [Todo] Advanced ESG/on-chain reporting.
- [Todo] Post-traction tokenomics design.

KPIs

Indicator	Target
TVL	50 M USD +
Countries	5 to 8
SMEs/actors financed	10,000 +
Farmers reached	100,000 +
Productive assets	250 +
Jobs supported	25,000 +

Gating decision

AfriCred becomes a pan-African productive finance infrastructure once the platform combines liquidity, measurable impact and regional distribution.

3. Governance, risks and safeguards

Risk	Recommended safeguard	Owner	Current status
Smart-contract risk	Full tests, Safe multisig, emergency pause, external audit before any cap increase	CTO / Blockchain	Tests [Done] ; slither clean [Done] ; multisig, pause, audit [Todo]
Liquidity risk	Clear lock-up, asynchronous redemptions, liquidity buffer	Treasury	Implicit lock-up via state ; synchronous redemptions in OpenWithdrawal ; no buffer
Portfolio risk	SME scoring, sector/country diversification, concentration limits, arrears monitoring	Risk committee	risk field (Low/Medium/High) on the LoanNFT; no scoring or concentration limits
Impact/data risk	Verifiable field data, allocation proofs, monthly then quarterly reporting	Impact / Operations	IPFS dossier [Done] ; allocation proofs [Todo] ; reporting [Todo]
Reputation risk	No guaranteed-yield promise, signed disclosure, prudent communication	Management / Legal	DisclaimerDialog [Done] ; signed disclosure [Todo]

4. Next 90-day execution plan (Phase 2 ramp)

Period	Priority	Actions	Expected output
Days 1-15	Phase 2 scoping	Size the Safe multisig (2/3 then 3/5), lock the pilot investor list via allow-list, sign off on the mainnet scope	Pilot build authorised
Days 16-35	Contract hardening	Add <code>Pausable</code> (OZ), <code>minDeposit</code> , <code>maxDepositPerWallet</code> , richer events for reporting	Contracts ready for audit
Days 36-55	External audit + fixes	Run the audit on Vault, Factory, LoanRegistryNFT, Router; fix findings; optional Immunefi/Cantina bug bounty	Clean audit report
Days 56-70	Monitoring + runbooks	Configure Tenderly (alerts on lifecycle transitions, custodyFunds, recordRecovery); write admin and treasury runbooks	Production-ready ops stack
Days 71-80	Mainnet deploy	Deploy Factory + Router on Base mainnet via the Safe; whitelist USDC; deploy the first pilot vault	MVP live on mainnet (closed allow-list)
Days 81-90	Private beta	Onboard pilot investors (5 to 10), first SME loan in production, first monthly report	First repayment recorded on-chain

5. Conclusion

The most credible path is to prove a simple, mastered loop first: **stablecoin on Base -> AfriCred vault -> productive SME allocation -> repayment -> impact reporting**. That loop is **validated on testnet today** via the AgriCo Senegal pilot vault at `0x027c0dA455141827a76B5Aa2Dda2e85FB222001C`. Phases 0 and 1 are closed.

To move into Phase 2 (private MVP on mainnet), two blocking technical workstreams remain:

1. **Security:** `Pausable` (emergency pause), Safe multisig on the factory, external audit.
2. **Operations:** on-chain monitoring (Tenderly), signed runbooks, mainnet Safe and deployment configuration.

Once those workstreams ship and the mainnet pilot is stable, AfriCred can accelerate toward productive assets, multi-country, multi-chain, mobile money and institutional capital.